

write below this line

Questions is to be answered on the Question Paper and the rest on the answer sheet.
Complete Question Paper is to be returned back along with the answer sheet.

Recognize the fundamental economic and financial considerations involved in engineering
in the blanks/select the best choice.

Which of the following methods is used to evaluate investment alternatives?

- a. Payback period
- b. Net present value (NPV)
- c. Internal rate of return (IRR)
- ☒ d. All of the above

Which of the following is a measure of the profitability of an investment?

- a. Payback period
- ☒ b. Return on investment (ROI)
- c. Discount rate
- d. Opportunity cost

What is the payback period?

- ☒ a. The time it takes to recoup the initial investment
- b. The time value of money
- c. The discount rate used in investment analysis
- d. The measure of project profitability

Which of the following is true about net present value (NPV)?

- ☒ a. A positive NPV indicates a profitable investment
- b. NPV is not influenced by the discount rate
- c. NPV is used to measure short-term financial gains
- d. A negative NPV indicates a profitable investment

**National University of Computer and Emerging Sciences
Lahore Campus**

1. Which of the following factors should be considered in engineering economic analysis?

- a. Initial investment cost
- b. Maintenance and operating costs
- c. Salvage value
- ☒ d. All of the above

Which of the following is not part of the opportunity cost of going on holiday?

- a. The money you spent on a theater show
- ☒ b. The money you could have made if you had stayed at home and worked
- c. The money you spend on airline tickets
- d. the money you spent on food

Opportunity cost is _____.

- a. the cost incurred in the past before we decide what to do in the future.
- b. a cost that cannot be avoided, regardless of what is done in the future
- ☒ c. that which we forgo, or give up, when we make a choice or a decision.
- d. the additional benefit of buying an additional unit of a product

Microeconomics is not concerned with the behavior of _____.

- ☒ a. aggregate demand
- b. firms
- c. Consumers
- d. industries

In economics, the central problem is _____.

- a. Scarcity
- ☒ b. Money
- c. Consumption
- d. Allocation

MARR is established by the accounts managers of the firm. (True / ☒ False)

The ROR, expressed as a percentage rate per year, is called the cost of capital. (True / ☒ False)

MARR usually considers the risk inherent to a project. (☒ True / False)

CPI measures the average change over time in the prices paid by consumers for goods and services serving as a key indicator of inflation. (☒ True / False)

Designing does not involve economic decisions. (True / ☒ False)

Designing does not involve economic decisions. (True / ☒ False)

Q1: Recognize the fundamental economic and financial considerations involved in engineering projects and briefly describe the three broad types of economic systems.

Q2: Recognize the fundamental economic and financial considerations involved in engineering projects and briefly describe the three broad types of economic systems.

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects
Q4: Using a flow chart, show the steps involved in an Engineering Economy study.

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects
Q5: If a company sets aside \$1,000,000 now into a contingency fund, how much will the company have in 5 years, if it does not use any of the money and the account grows at a rate of 10% per year?

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects
Q6: Managers from different departments in Zenith Trading, a large multinational corporation, have offered six projects for consideration by the corporate office. A staff member for the chief financial officer used the following keywords to identify the projects and then listed them in order of projected rate of return, as shown below. If the company wants to grow rapidly through high leverage and uses only 10% equity financing, that is, a cost of equity capital of 9% and 90% debt financing with a cost of debt of 5% (Assumption).

Project ID	Projected ROR. % per year
Inventory	30
Technology	28.4
Warehouse	19
Products	13.1
Energy	9.6
Shipping	8.2